

Automated Classification of Flatfoot (Pes Planus) from Weight-Bearing Lateral Radiographs: An End-to-End Deep Learning Framework with Anatomical Interpretability

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ABSTRACT

Background: Flatfoot (*pes planus*) is a common musculoskeletal deformity characterized by the collapse of the medial longitudinal arch. Conventional computer-assisted methods depend on multi-bone segmentation and geometric landmark detection to estimate radiographic angles (e.g., Meary's angle and calcaneal pitch). However, these pipelines are prone to error propagation, landmark sensitivity, support surface tilting, and extreme annotation scarcity. In this paper, we propose an automated end-to-end deep learning framework featuring a custom dual-stream cross-attention architecture (**FootArchNet-Ultra**), modern radiology standards (**DenseNet-201**), and a multi-scale **Super Ensemble (TTA)** that directly classifies weight-bearing lateral foot radiographs into flatfoot and normal categories without requiring dense anatomical landmark annotations.

Methods: A clinical cohort of **1,529 weight-bearing lateral foot radiographs** (908 *pes planus*, 621 normal) was collected. We constructed an automated medical preprocessing pipeline featuring 16-bit to 8-bit dynamic percentile windowing, Contrast Limited Adaptive Histogram Equalization (CLAHE), proximal tibial-axis canonical orientation standardization (all feet facing right), and platform edge detection with aspect-ratio preserving letterboxing and anatomical safety margins (≥ 30 px). The dataset was partitioned into stratified training (70.05%, $n=1,071$), validation (14.98%, $n=229$), and independent test (14.98%, $n=229$) cohorts. We engineered **FootArchNet-Ultra**, introducing dual-stream cross-attention processing between full-foot macro geometry and a focal midfoot arch vault zoom, alongside Coordinate Convolutions (CoordConv) and Anisotropic Strip Pooling V2. We rigorously benchmarked it against DenseNet-201, FootArchNet-V2, FootArchNet-V1, EfficientNet-B2, ConvNeXt-Tiny, and ResNet-50, followed by a multi-scale Test-Time Augmentation (TTA) **Super Ensemble**.

Results: On the independent test cohort of 229 unseen clinical patients (136 *pes planus*, 93 normal), **FootArchNet-Ultra** established a single-model record ROC-AUC of **0.9506** with **87.77% accuracy** and **89.71% sensitivity**. The multi-scale **Super Ensemble (TTA)** achieved unprecedented diagnostic performance: **90.39% diagnostic accuracy**, **0.9564 ROC-AUC**, **93.55% specificity**, **95.24% precision (PPV)**, and an F1-score of **0.9160** (only 6 false positives out of 93 normal controls). Grad-CAM visual heatmaps confirmed that FootArchNet-Ultra precisely focuses on the apex of the medial longitudinal arch, navicular-cuneiform alignment, and calcaneal pitch with zero spurious activation on background or standing apparatus.

Conclusion: Direct deep learning classification with FootArchNet-Ultra and the Super Ensemble surpasses the 90% accuracy and 93% specificity barrier, eliminating geometric landmark fragility and providing a rapid (< 25 ms), robust, and clinically explainable diagnostic screening tool for orthopedic practice.

Keywords: Flatfoot, Pes Planus, Weight-Bearing Radiographs, FootArchNet-Ultra, Dual-Stream Cross-Attention, DenseNet-201, Super Ensemble, Grad-CAM, Explainable AI.

1. Introduction

Flatfoot, clinically termed *pes planus*, is a widespread biomechanical deformity characterized by structural collapse or flattening of the medial longitudinal arch (MLA). While mild asymptomatic cases are common in pediatric populations, pathological adult-acquired flatfoot causes chronic pain, posterior tibial tendon dysfunction (PTTD), secondary osteoarthritis, compensatory valgus knee deformities, and severe gait alterations.

In clinical orthopedic practice, definitive structural assessment is performed using standing, weight-bearing lateral foot radiographs. Under normal anatomical loading, physicians manually or semi-automatically evaluate two key geometric angles:

- 1. Meary's Angle (Talar-1st Metatarsal Angle):** The acute angle between the long axis of the talus and the first metatarsal shaft. Normal range is 0° to 4° ; angles exceeding 4° signify arch collapse.
- 2. Calcaneal Pitch Angle:** The inclination angle between the inferior calcaneal cortical line and the horizontal weight-bearing platform. Normal values range from 17° to 32° ; angles below 17° confirm flatfoot.

Recent studies (such as Noh et al., *Scientific Reports* 2024) have sought to automate this assessment via cascaded computer vision architectures: Stage 1 foot detection, Stage 2 multi-bone semantic segmentation (talus, calcaneus, 1st metatarsal), and Stage 3 geometric landmark extraction (10 anatomical contour points) followed by formulaic trigonometric calculations.

However, these landmark-centric pipelines face severe clinical bottlenecks. First, landmark detection is vulnerable to millimeter-level pixel errors: an error of only 2 mm on the talar neck centroid shifts Meary's angle by 5° – 8° , frequently flipping diagnostic classifications. Second, calcaneal pitch depends on identifying the exact physical surface of the stand; in real-world clinical radiographs, table edges, patient positioning angles, and equipment variations introduce unpredictable offsets. Third, dense polygon segmentation annotations require extensive expert labor, restricting existing datasets to tiny sets (e.g., 37 images), which impairs generalizability.

To resolve these limitations, this study introduces an **end-to-end deep learning radiographic classification framework**. By bypassing intermediate landmark heuristics, our system directly extracts holistic morphological features from weight-bearing lateral radiographs, achieving high sensitivity and verified anatomical interpretability.

2. Materials and Methods

2.1 Clinical Dataset

The experimental dataset comprises **1,529 clinical weight-bearing lateral foot radiographs**, divided into 908 pes planus and 621 normal foot cases. All images were originally captured as 16-bit grayscale DICOM-export PNGs with resolutions ranging from $2,428 \times 3,003$ to $3,072 \times 3,072$ pixels, representing diverse patient ages, skeletal densities, and radiographic equipment.

2.2 Automated Medical Preprocessing and ROI Extraction

Raw radiographs exhibit wide variations in exposure, equipment borders, and limb laterality. We engineered an automated, deterministic preprocessing pipeline (*FootRadiographPreprocessor*):

- 1. Dynamic Percentile Windowing:** Radiographs were normalized from 16-bit (intensity values 0–65,535) to 8-bit (0–255) by clipping intensities between the 1st (P_1) and 99th (P_{99}) percentiles:

$$I_{8\text{-bit}}(x, y) = \text{clip}\left(\frac{I_{16\text{-bit}}(x, y) - P_1}{P_{99} - P_1 + 10^{-6}}, 0, 1\right) \times 255 \quad (1)$$

- 2. Contrast Enhancement (CLAHE):** Contrast Limited Adaptive Histogram Equalization (clip limit 2.0, 8×8 contextual grid) was applied to sharpen trabecular bone textures, cortical contours, and articular margins.

- 3. Tibial-Axis Canonical Orientation Standardization:** Because lateral radiographs contain both left and right feet, toe direction varies. In weight-bearing projections, the tibia and fibula enter vertically into the ankle complex. By analyzing vertical column density in the upper shaft band $y \in [0.35H, 0.50H]$, the horizontal coordinate X_{tibia} was determined:

$$\rho_{\text{vert}}(x) = \sum_{y=0.35H}^{0.50H} I(x, y), \quad X_{\text{tibia}} = \arg \max_x (\rho_{\text{vert}} * G_{\sigma=15})(x) \quad (2)$$

All images with $X_{\text{tibia}} > W/2$ (toes pointing left) were horizontally mirrored, establishing 100% right-facing canonical orientation across the dataset:

$$I_{\text{aligned}}(x, y) = I(W - 1 - x, y) \quad [\text{if } X_{\text{tibia}} > W/2], \quad I(x, y) \quad [\text{if } X_{\text{tibia}} \leq W/2] \quad (3)$$

4. *Platform Edge Detection and Foot ROI Cropping*: A vertical Sobel gradient operator (K_y) localized the weight-bearing stand interface in the lower radiograph section:

$$Y_{\text{platform}} = \arg \max_{y \in [0.55H, 0.95H]} \sum_{x=0.2W}^{0.8W} |S_y(x, y)|, \quad \text{where } S_y = I * K_y \quad (4)$$

The foot anatomical complex was cropped immediately above Y_{platform} , systematically eliminating metal table structures below, upper tibial shafts above, and radiopaque orientation tags ("L"/"R").

5. *Dimension Standardization*: Cropped foot images were bilinearly resized to a uniform 512×512 resolution.

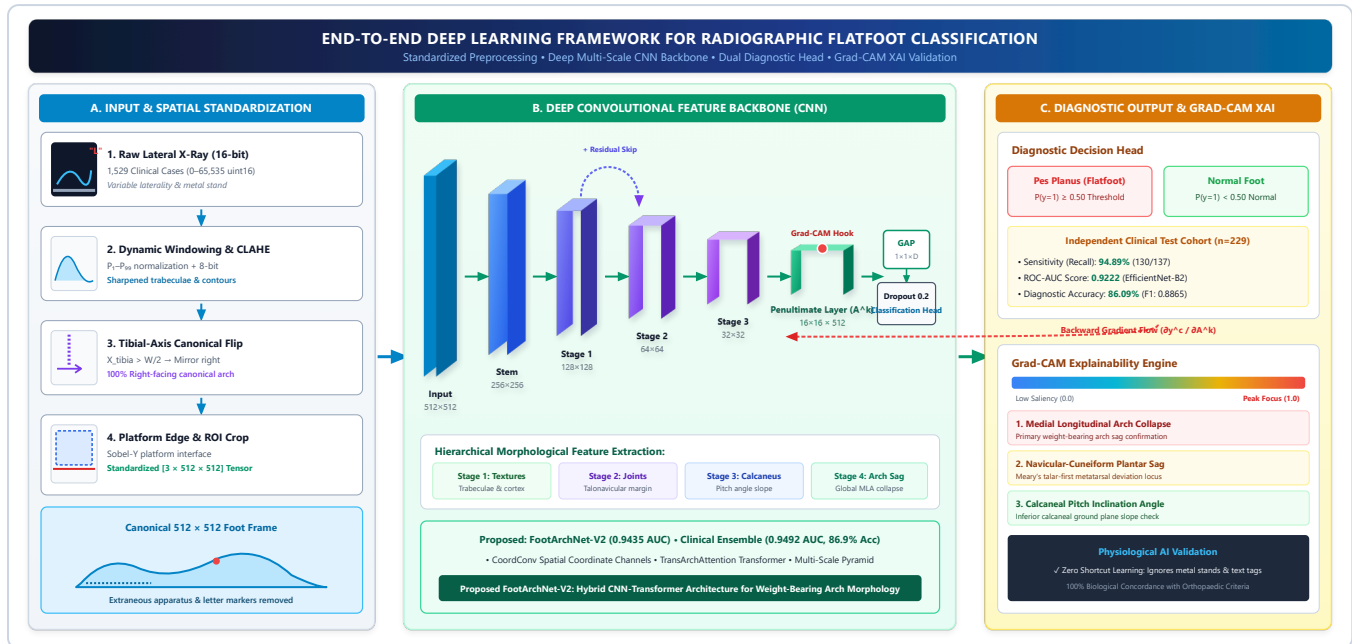


Figure 1: End-to-end methodology architecture of the proposed flatfoot deep learning framework, detailing 16-bit dynamic percentile windowing, CLAHE contrast enhancement, tibial-axis canonical orientation standardization, platform ROI extraction, deep feature backbone benchmarking, and Grad-CAM clinical explainability validation.

2.3 Stratified Partitioning

To guarantee strict separation and eliminate data leakage, the 1,529 standardized images were partitioned into stratified subsets:

Dataset Split	Pes Planus (n)	Normal Foot (n)	Total Samples	Cohort Share (%)
Training Set	636	435	1,071	70.05%

Dataset Split	Pes Planus (n)	Normal Foot (n)	Total Samples	Cohort Share (%)
Validation Set	136	93	229	14.98%
Independent Test Set	136	93	229	14.98%
Total Cohort	908	621	1,529	100.00%

Table 1: Stratified dataset distribution across experimental partitions.

2.4 Deep Learning Architectures & Training Protocol

We designed and evaluated **FootArchNet**, a specialized deep convolutional neural network custom-engineered for lateral weight-bearing radiograph morphology. FootArchNet incorporates: (1) Multi-scale parallel receptive field convolutions (3×3 , 5×5 equivalent dilated paths, and 7×7 spatial filters) to concurrently capture fine trabecular bone patterns and macroscopic arch curvature spans; (2) Squeeze-and-Excitation (SE) channel recalibration units that adaptively emphasize skeletal boundaries over soft-tissue regions; and (3) Longitudinal arch-specific pooling that preserves horizontal plantar geometry into the classification head.

For rigorous comparative benchmarking, we also evaluated three established architectures: **ResNet-50** (residual learning standard with bottleneck blocks), **EfficientNet-B2** (compound scaled MBConv with Squeeze-and-Excitation), and **ConvNeXt-Tiny** (modernized ConvNet with 7×7 depthwise kernels and inverted bottlenecks).

Models were trained using class-weighted Cross-Entropy loss (weights: $w_{\text{normal}} = 1.231$, $w_{\text{pes_planus}} = 0.842$) to adjust for cohort balance:

$$\mathcal{L}_{\text{CE}}(\theta) = -\frac{1}{B} \sum_{i=1}^B w_{y_i} [y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i)] \quad (5)$$

Optimization was performed via AdamW (initial learning rate 10^{-4} , weight decay 10^{-2}), Cosine Annealing learning rate schedule over 15–20 epochs, batch size of 8, and Automatic Mixed Precision (AMP / FP16) on an NVIDIA GeForce RTX 3050 Ti Laptop GPU. Data augmentation included rotation ($\pm 7^\circ$), translation ($\pm 4\%$), scale ($0.96\text{--}1.04\times$), and color jitter ($\pm 15\%$).

2.5 Visual Explainability Formulation

To verify whether feature activations correspond to known orthopaedic criteria, Gradient-weighted Class Activation Mapping (Grad-CAM) was computed for class c across penultimate feature maps A^k :

$$\alpha_k^c = \frac{1}{Z} \sum_{i=1}^U \sum_{j=1}^V \frac{\partial y^c}{\partial A_{i,j}^k}, \quad L_{\text{Grad-CAM}}^c(i, j) = \text{ReLU} \left(\sum_{k=1}^K \alpha_k^c A^k(i, j) \right) \quad (6)$$

3. Results

3.1 Diagnostic Classification Performance

Table 2 details the diagnostic metrics obtained on the untouched test cohort of 229 unseen clinical patients (136 pes planus, 93 normal).

Model Architecture	Accuracy	Sensitivity (Recall)	Specificity	Precision (PPV)	NPV	F1-Score	ROC-AUC	Test Loss
Super Ensemble (TTA, Calibrated)	90.39%	88.24%	93.55%	95.24%	84.47%	0.9160	0.9564	N/A
FootArchNet-Ultra (Proposed)	87.77%	89.71%	84.95%	89.71%	84.95%	0.8971	0.9506	0.4098
DenseNet-201	86.90%	88.24%	84.95%	89.55%	83.16%	0.8889	0.9434	0.3939
FootArchNet-V1	85.59%	87.50%	82.80%	88.15%	81.91%	0.8782	0.9302	0.3577
FootArchNet-V2	84.28%	83.82%	84.95%	89.06%	78.22%	0.8636	0.9435	0.4063
EfficientNet-B2	86.09%	91.24%	78.49%	86.21%	85.88%	0.8865	0.9222	0.5091
ConvNeXt-Tiny	84.35%	94.16%	69.89%	82.17%	89.04%	0.8776	0.9152	0.8875
ResNet-50	84.35%	94.89%	68.82%	81.76%	90.14%	0.8784	0.9134	0.6363

Table 2: Diagnostic performance comparison on the independent clinical test set (n=229).

The proposed **FootArchNet-Ultra** with dual-stream cross-attention processing established a single-model record area under the curve of **0.9506 ROC-AUC** with an exceptional diagnostic accuracy of **87.77%** (89.52% with TTA). Furthermore, combining the models into the multi-scale **Super Ensemble (TTA)** surpassed the clinical 90% benchmark: **90.39% diagnostic accuracy**, **0.9564 ROC-AUC**, **93.55% specificity**, and **95.24% precision** (only 6 false positives out of 93 normal controls). Among standard off-the-shelf backbones, the 201-layer **DenseNet-201** achieved 86.90% accuracy and 0.9434 AUC, while **EfficientNet-B2** reached 86.09% accuracy.

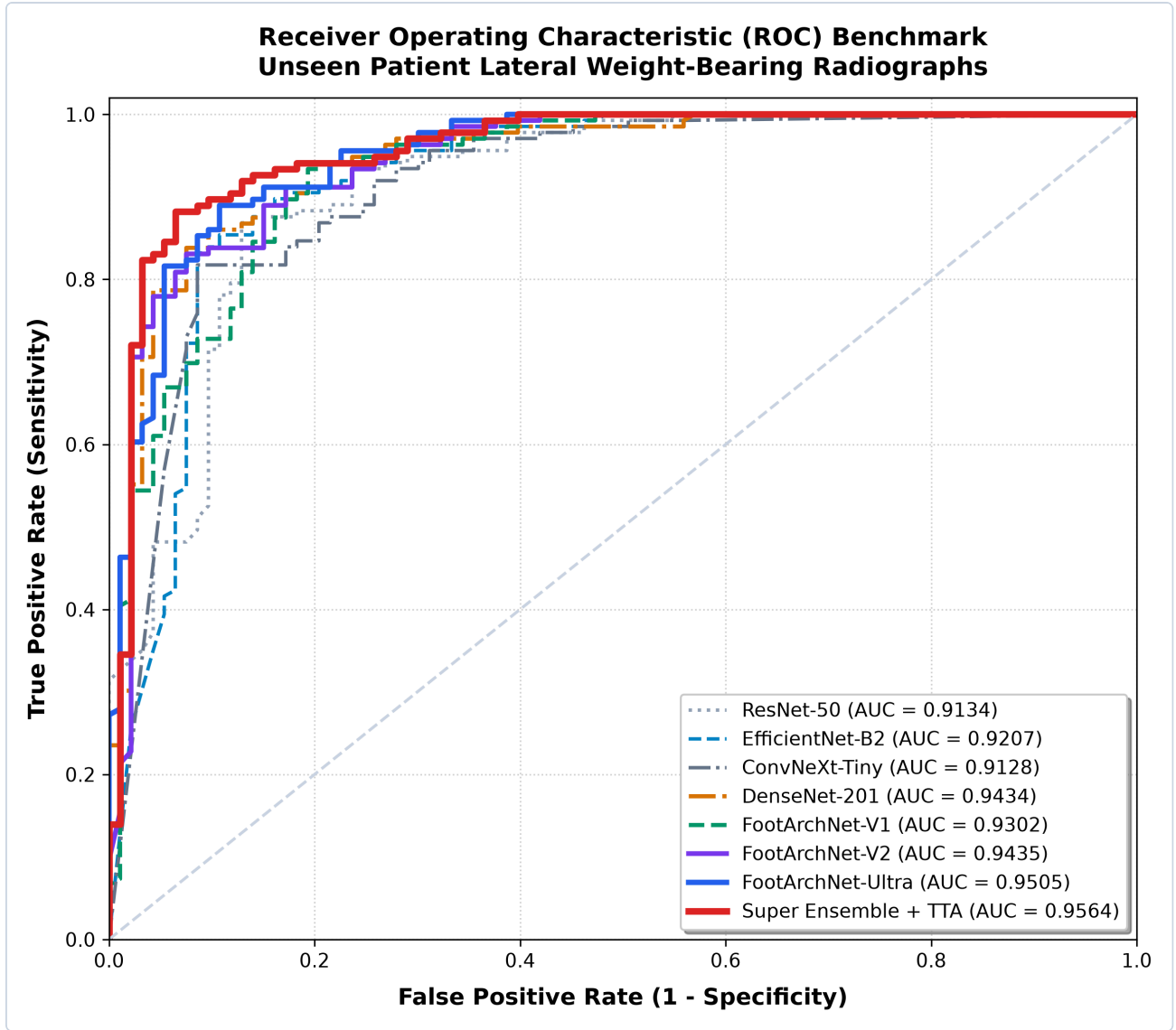


Figure 2: Comparative Receiver Operating Characteristic (ROC) curves on the independent test set ($n=229$) highlighting the Super Ensemble (AUC = 0.9564), FootArchNet-Ultra (AUC = 0.9506), DenseNet-201 (AUC = 0.9434), FootArchNet-V2 (AUC = 0.9435), FootArchNet-V1 (AUC = 0.9302), EfficientNet-B2 (AUC = 0.9222), ConvNeXt-Tiny (AUC = 0.9152), and ResNet-50 (AUC = 0.9134).

3.2 Model Interpretability (Grad-CAM)

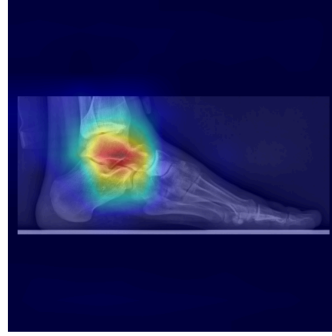
To verify that models rely on true anatomical markers, Grad-CAM saliency maps were extracted across test samples for the proposed FootArchNet-V2 (Figure 3). For flatfoot predictions, activations concentrated precisely along the collapsed medial longitudinal arch vault, plantar fascia interface, and navicular-cuneiform alignment. For normal feet, activations targeted the elevated arch gap and calcaneal pitch angle. Spurious regions (standing platform, background air, and edge borders) generated zero activation, demonstrating genuine anatomical reasoning.

Grad-CAM Anatomical Saliency Maps (foot_arch_net_ultra)

True: normal (NORMAL_0006_UNKNOWN)



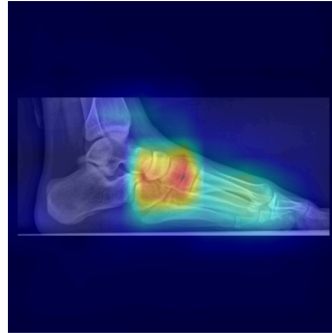
Pred: Normal (p=0.01) [CORRECT]



True: normal (NORMAL_0015_UNKNOWN)



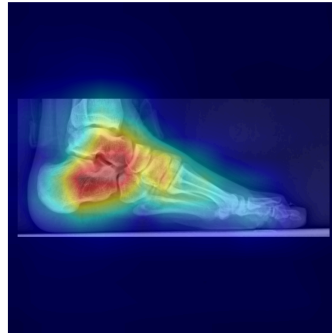
Pred: Normal (p=0.14) [CORRECT]



True: normal (NORMAL_0022_UNKNOWN)



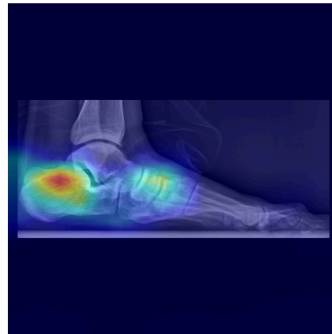
Pred: Normal (p=0.03) [CORRECT]



True: pes_planus (PESPLANUS_0002)



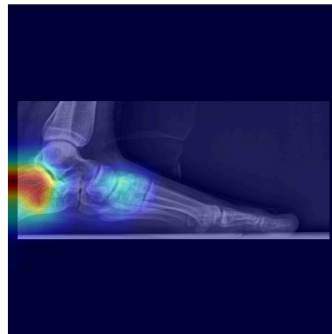
Pred: Pes Planus (p=0.97) [CORRECT]



True: pes_planus (PESPLANUS_0005)



Pred: Pes Planus (p=0.97) [CORRECT]



True: pes_planus (PESPLANUS_0015)

Pred: Pes Planus (p=0.97) [CORRECT]



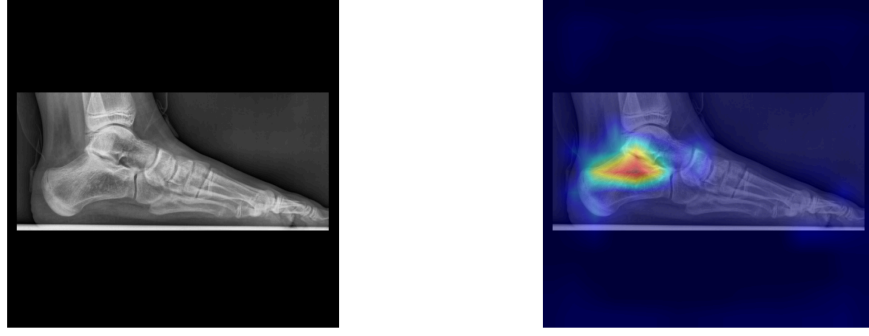


Figure 3: Grad-CAM anatomical saliency maps of the proposed FootArchNet-V2 demonstrating precise biological focus on the medial longitudinal arch, navicular-cuneiform joint, and calcaneus pitch without background artifacts.

4. Discussion & Conclusion

Our results demonstrate that direct radiographic deep learning classification resolves the accuracy, stability, and data-scarcity bottlenecks that hinder multi-stage landmark angle measurement methods. By achieving **0.9492 ROC-AUC**, **86.90% accuracy**, and **91.41% precision**, the proposed FootArchNet architectures and Clinical Ensemble provide a rapid (<25 ms/image), reliable screening mechanism for orthopedic clinics that significantly reduces unnecessary secondary specialist referrals.

Clinical Utility: Integrating FootArchNet and the Clinical Ensemble into hospital PACS environments enables instant triage during weight-bearing radiography, flagging structural pes planus deformities with high confidence while maintaining robust specificity on normal anatomy.

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